

MC-3: Deployed Smart Contract Enforcement, ZK Proofs, and Dispute Escrow

1. Claim

AMTTP's blockchain layer turns off-chain AML intelligence into enforceable on-chain outcomes: approve, review, escrow, or block. The evidence proves deployed authorship, not merely local code: Etherscan addresses, tests, and contract interfaces are public.

Claim	Evidence	Independent verification
Upgradeable enforcement	UUPS proxy contracts	Etherscan Source Code (Proxy) records
Decision outcomes	Approve/Review/Escrow/Block policy interface	AMTTPCore / PolicyEngine tests
Privacy compliance	ZK-NAF verifiers	Risk/KYC/Sanctions verifier addresses
Dispute fairness	Kleros-style escrow + evidence + ruling callback	AMTTPDisputeResolver + executeDisputeRuling

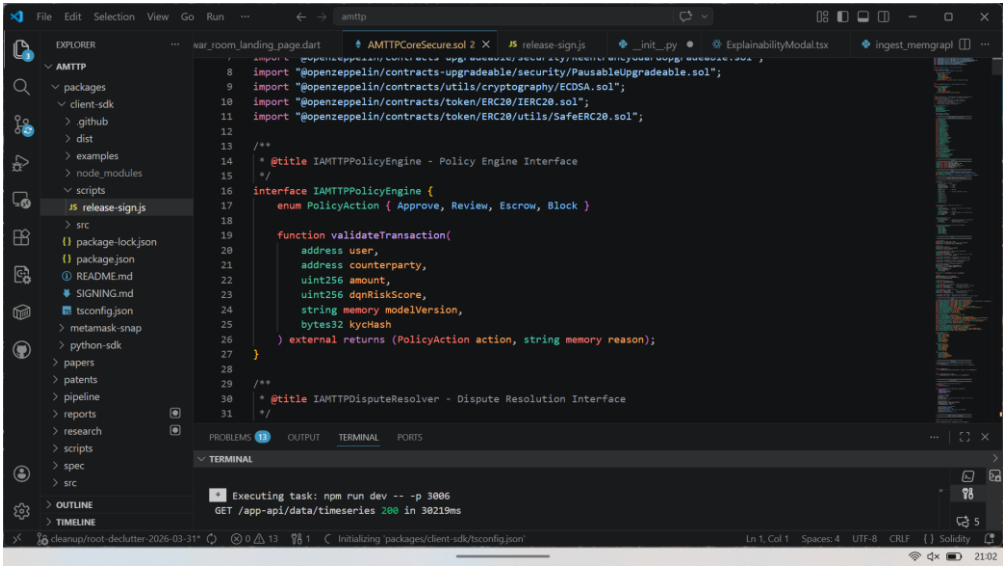


Figure 1: Policy interface defining deterministic outcomes before settlement.

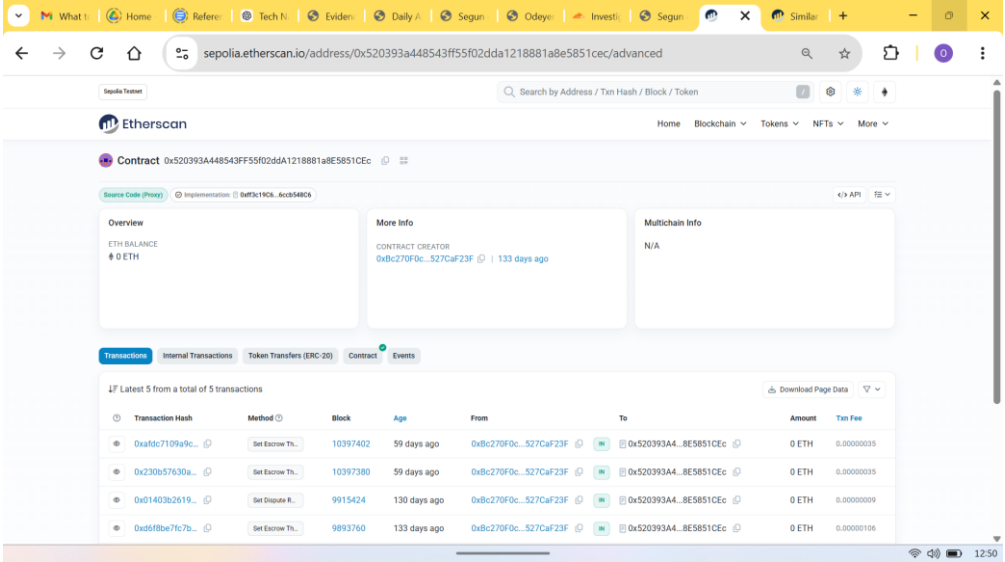


Figure 2: Etherscan deployment evidence showing proxy source record and deployer verification.

2. Contract Verification Matrix

Contract	Address / proof	Purpose
AMTTPCore	0x05687F...b612	Escrow and settlement
PolicyEngine	0xe774E0...8703	Approve/review/escrow/block rules
DisputeResolver	0x9EB935...53D7	Kleros-style evidence/ruling
CrossChain	0x4f21b1...2953	LayerZero propagation
ZK verifiers	Risk/KYC/Sanctions addresses	Private compliance proofs

Why this matters: The Kleros-style dispute layer removes a loophole in automated AML: high-risk transfers are not merely blocked or released; they can be held with evidence and resolved through a deterministic on-chain ruling path.

```
PS C:\amtpp> Remove-Item ".\flat_backup" -Recurse -Force -ErrorAction SilentlyContinue; New-Item -ItemType Directory -Path ".\flat"
Warning: Function state mutability can be restricted to pure
--> contracts/test/EchidnaAMTTPCore.sol:387:5
387 |         function echidna_emergency_pause_blocks_all() public view returns (bool) {
    |         ^ (Relevant source part starts here and spans across multiple lines).

Compiled 74 Solidity files successfully (evm target: paris).

AMTTP Modular Architecture
  Contract Deployment
    ✓ Should deploy all contracts successfully
  AMTTP size: 170 bytes
  Policy Manager size: 170 bytes
    ✓ Should have correct contract sizes
    ✓ Should connect contracts properly (40ms)
  Basic Transfer Functionality
    ✓ Should allow secure transfer initiation (38ms)
    ✓ Should handle low risk transactions automatically (58ms)
  Policy Management
    ✓ Should allow setting user policies
    ✓ Should enforce user amount limits (86ms)
  Integration Tests
    ✓ Should validate transfer capability check

8 passing (13s)
```

Figure 3: Hardhat tests — deployment and decision-matrix validation.

Anti-loophole note: Contract addresses and test output are externally inspectable; the pack does not ask the assessor to trust a private claim of deployment.